DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

Second Semester, SY 2024-2025

COURSE CODE	**:**	S-CSIS426
COURSE TITLE	**:**	Special Topics in Artificial Intelligence
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two units
MODALITY	**:**	Traditional blended learning model
PREREQUISITE	**:**	None
PROFESSOR	:	Ms. Jennylinde R. Manaois
CONSULTATION HOURS	:	M / 12:30 - 1:00 / online, T / 8:00 - 12:00 / online, H / 1:00 - 2:30 / COS100c, F / 8:00-12:00 / online
CONTACT DETAILS	:	jrmanaois@dlsud.edu.ph

VISION:

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION:

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION

Deals with topics on specialized or current interest which may change. It includes expert systems, natural language processing, neural network models, machine learning, and AI applications in remote sensing.

COURSE LEARNING OUTCOMES

By the end of this course, students can:

CLO1. relate a deeper understanding of the history of artificial intelligence (AI) and its foundations;
CLO2. understand intelligent agents and their operational contexts;
CLO3. apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning;
CLO4. recognize the concepts of artificial intelligence in different fields;
CLO5. discover the importance of technology by creating different applications; and
CLO6. relate 21st century skills in all learning activities.

LEARNING PLAN

Learning Outcomes for Midterm Period;

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Relate a deeper understanding of the history of artificial intelligence (AI) and its foundations.	• TLO1. Distinguish the history of AI and its foundations.
CLO2. Understand intelligent agents and their operational contexts.	• TLO2. Classify the structures of agents and its environments. • TLO3. Differentiate between various types of intelligent agents (e.g., simple reflex agents, model-based agents, goal-based agents, utility-based agents). • TLO4. Describe different types of environments in which intelligent agents operate (e.g., fully observable vs. partially observable, deterministic vs. stochastic, episodic vs. sequential). • TLO5. Explain the architecture of intelligent agents, including the perception, reasoning, and action components.
CLO3. Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.	• TLO6. Define problem solving in the context of artificial intelligence. • TLO7. Differentiate between various types of problems (e.g., deterministic vs. stochastic, single-agent vs. multi-agent). • TLO8. Describe different search strategies used in AI, including uninformed (blind) search and informed (heuristic) search. • TLO9. Compare and contrast various search algorithms such as breadth-first search, depth-first search, uniform-cost search, and A* search.

Module No and Learning Outcomes	**Date**	**Teaching-Learning Activities / Assessment Strategy**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**
Module 0 TLO15 CLO 6	Week 1 (Jan.20-24)	• Gospel reading, Reflection and Prayer • Discussion of Syllabus • Discussion of Project	PowerPoint, Multimedia Devices (Laptop, Projector)	/		2

Module 1 TLO 1 CLO 1	Week 2 (Jan.27-Jan.31)	• Gospel reading, Reflection and Prayer • Module 1: Introduction to AI • Enabling Assessment: Assignment on Introduction to AI (10 pts.)	Schoolbook MS Teams		/	2
Module 1 TLO 1 CLO 1	Week 3 (Feb.3-7)	• Gospel reading, Reflection and Prayer • Module 1: Introduction to AI • Enabling assessment: Short Quiz on Introduction to AI (20 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)	/		2
Module 2 TLO 2,3,4,5 CLO 2	Week 4 (Feb.10-14)	• Gospel reading, Reflection and Prayer • Module 2: Intelligent Agents • Enabling assessment: Assignment on Intelligent Agents (30 pts.)	Schoolbook		/	2
Module 1,2 TLO 1,2,3,4,5 CLO 1,2	Week 5 (Feb.17-21)	• Long quiz on Module 1 and 2 (40 pts.)	Schoolbook	/		2
Module3 TLO 6,7,8,9 CLO 3	Week 6 (Feb.24-Feb.28)	• Gospel reading, Reflection and Prayer • Module 3: Problem Solving and Search Strategies	Schoolbook		/	2
Module3 TLO 6,7,8,9 CLO 3	Week 7 (Mar.3-7)	• Gospel reading, Reflection and Prayer • Module 3: Problem Solving and Search Strategies • Enabling assessment: Short Quiz on module 3 (20 pts.)	Schoolbook MS Teams	/		2
Module3 TLO 6,7,8,9,15 CLO 3,6	Week 8 (Mar.10-14)	• Review of Problem Solving and Search Strategies • Research RRL • Submission of RRL	Schoolbook MS Teams		/	2
Module 1,2,3 TLO 1,2,3,4,5, 6,7,8,9 CLO 1,2,3	Week 9 (Mar.17-21)	**MIDTERM EXAMINATION** Written Examination covering Modules 1-3		/		2
			TOTAL	***10.0***	***8.0***	***18.0***

Learning Outcomes for Final Period;

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Recognize the concepts of artificial intelligence in different fields.	• TLO10. Apply different techniques to create games and applications. • TLO11. Classify learning from examples, knowledge in learning and reinforcement learning. • TLO12. Apply genetic algorithm to solve optimization problems. • TLO13. Discover robotic software architectures.
CLO5. Discover the importance of technology by creating different applications.	• TLO14. Apply the different topics you learned in AI to come up with a new innovation.
CLO6. Demonstrate 21st century skills in all learning activities.	• TLO15. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World ProblemSolving, Use of ICT (not part of topic, but integrated in assessments)

Module No and Learning Outcomes	**Date**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**		**Asynchronous**	**Allotted Hours**
Module 4 TLO 10 CLO 4	Week 10 (Mar.24-28)	• Gospel reading, Reflection and Prayer • Module 4: Game Playing and Adversarial Search • Sustainability Week	PowerPoint, Multimedia Devices (Laptop, Projector)	/		2
Module 5 TLO 11 CLO 4	Week 11 (Mar.31-Apr.4)	• Gospel reading, Refelction and Prayer • Module 5: Artificial Neural Network • Enabling assessment: Short Quiz on Artificial Neural Network (10 pts.)	Schoolbook		/	2
Module 6 TLO 12 CLO 4	Week 12 (Apr.7-11)	• Gospel reading, Reflection and Prayer • Module 6: Introduction to Evolutionary Computing • Enabling assessment: Short Quiz on Evolutionary Computing	PowerPoint, Multimedia Devices (Laptop, Projector)	/		2

	Week 13 (Apr.14-18)	Independent Learning Week			/	2
Module 7 TLO 13 CLO 4	Week 14 (Apr.21-25)	• Gospel reading, Reflection and Prayer • Module 7: Robotics and Robot Motion Planning • Research Week	Schoolbook		/	2
Module 4,5,6,7 TLO 11,12,13 CLO 4	Week 15 (Apr.28-May2)	• Gospel reading, Reflection and Prayer • Long Quiz on Module 4, 5, 6 and 7 (40 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)	/		2
Module 1,2.3.4,5,6,7 TLO 14,15 CLO 5,6	Week 16 (May5-9)	Summative assessment: Presentation of Project		/		2
			TOTAL	***8.0***	***6.0***	***14.0***

FINAL PRODUCT/PERFORMANCE:

Look for software/programs or other examples of applications displaying Artificial Intelligence. Analyze its components, functionalities and how it is developed. Create your own software applying Artificial Intelligence.

GRADING SYSTEM

Midterm;

Enabling Assessments - 50%
Attendance/Class Participation - 20%
Summative Assessment - 30%
100%

Finals;

Enabling Assessments - 50%
Attendance/Class Participation - 20%
Summative Assessments - 30%
100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.

d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Online References;

E-provider	References
Online	Builtin. 2025. Artificial Intelligence. Built In. Available from: https://builtin.com/artificialintelligence
Online	SAS. 2025. Artificial Intelligence in Today's World. SAS. Available from: https://www.sas.com/en_us/insights/analytics/what-is-artificialintelligence.html
Online	TechTarget. A guide to artificial intelligence in the enterprise. SearchEnterpriseAI. Available from: https://www.techtarget.com/searchenterpriseai/definition/AI-Artificial-Intelligence
Online	European Parliament. 2020. Artificial Intelligence. Available from: https://www.europarl.europa.eu/RegData/etudes/STUD/2020/641547/EP_RS_STU(2020)641547_EN.pdf

Journals;

E-provider	References
Journal	Siam SI, Ahn H, Liu L, Alam S, Shen H, Cao Z, Shroff N, Krishnamachari B, Srivastava M, Zhang M. 2024 Aug 30. Artificial Intelligence of Things: A Survey. ACM Transactions on Sensor Networks. Available from: https://doi.org/10.1145/3690639
Journal	Ullah N, Khan JA, Falco ID, Sannino G. 2024 Nov 23. Explainable Artificial Intelligence: Importance, Use Domains, Stages, Output Shapes, and Challenges. ACM Computing Surveys. Available from: https://doi.org/10.1145/3705724
Journal	Melder T, Savery R. 2024 Sep 18. Artificial Intelligence and Home Music Listening: An Overview and Future Directions. Audio Mostly 2024 - Explorations in Sonic Cultures. p. 318–324. Available from: https://doi.org/10.1145/3678299.3678330
Journal	Sun Q, Akman A, Schuller BW. 2024 Dec 21. Explainable Artificial Intelligence for Medical Applications: A Review. ACM Transactions on Computing for Healthcare. Available from: https://doi.org/10.1145/3709367

On-Site References;

Accession Number	References
AEA Lib Q 335 .Ar78 2022	Kaliraj P, Devi T. 2022. Artificial Intelligence: Theory, Models, and Applications. Boca Raton: CRC Press, Taylor & Francis Group.

AEA Lib Q 335 .L962 2022	Lucci S, Musa S, Kopec D. 2022. Artificial Intelligence in the 21st Century. Duxbury: Mercury Learning and Information..
AEA Lib Q 334.7 .C152 2022	Campbell C. 2022. AI by Design: A Plan for Living with Artificial Intelligence. Boca Raton: CRC Press.
AEA Lib Q 335 .Ar7 2019	Stirling K. 2019. Artificial Intelligence. New York (NY): Clanrye International.

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2nd Semester 2024-2025



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